

PROTEUS For Electronic Design Automation

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This generation (Gen Y, born between the 1980's and the year 2000) constantly gets bored. It is forever looking for something new and exciting, and everyone is willing to try different things as long as they can derive a good experience out of it. Be it in any field, it is difficult to stand out today simply because of the problem of plenty. Looking at the website of Proteus, I must say they are taking this aspect seriously.

Proteus design suite is a package put together by Labcenter Electronics to give you a novel simulation and printed circuit board (PCB) designing experience. What makes this tool special? What new does it have to offer? Read on to know.

Proteus design kit has about 785 microcontroller variants ready for simulation, straight from the schematic. Be it Atmel, Microchip or ARM, once you get a hang of the tool, you need not worry about the impending platform. Let us begin by considering one of the most popular microcontroller kits among students, Arduino.

Visual Designer for Arduino

Visual Designer in Proteus takes a cue from Arduino itself, removing the lower-level complexities of software language and hardware design. It uses a combination of its trademark Virtual System Modeling simulation—which we will come to later—flowcharts and virtual hardware, resulting in an integrated and intuitive development environment for Arduino.

Depending on the device you want to connect to Proteus, you can configure the peripheral accordingly and use high-level methods to control it. If you are attaching, say, a robot, you might

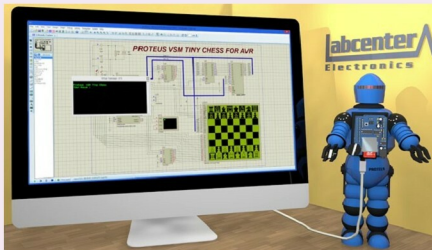


Fig. 1: Visual Designer for Arduino

Standard measuring instruments with Visual Designer

- Fully-featured four-channel digital storage oscilloscope with measurement cursors
- Counter/timer instrument for measuring time intervals and signal frequency
- DC and AC voltmeters and ammeters
- Dual-mode (master/slave) serial peripheral interface protocol analyser
- Pattern generator for creating 8-bit by 1kb digital data streams
- 40-channel logic analyser with capture buffer and measurement cursors
- RS232 terminal with configurable baud rate, data/stop bits and polarity
- Dual-mode (master/slave) inter-integrated circuit protocol analyser
- Signal generator to inject square, saw-tooth, triangle and sine waves
- An advanced simulation feature that lets you perform analysis on graphs, and even conformal analysis

need different peripherals for the various sensors, one for the wheels and another for the motor. The corresponding microcontroller is automatically updated with connection details.

Visual Designer allows you to set breakpoints at which you need to debug the course of the robot and perform simulations around it. To help you get started, the kit provides ready-to-use Arduino shields, breakout boards and sensors in its peripheral gallery. Transferring the data to Arduino board simply requires plugging in the programming

cable, making the required configurations and clicking to transfer the flowchart into the board. PICs can be simulated in real time, letting you test circuit functionality effectively.

Shortened lifecycle from schematic to PCB

It is VSM simulation that brings in Agile development into the embedded workflow, enabling you to quickly put together a prototype with hardware and firmware components. The inbuilt processor model is built to fully simulate input/output ports,